Dell PowerMax File

Protocol Guide

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DELL EMC

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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PREFACE

As part of an effort to improve its product lines, revisions of the software and hardware are periodically released. Therefore, some functions described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

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NOTE: This document was accurate at publication time. Go to Dell Online Support (https://support.emc.com) to ensure that you are using the latest version of this document.

Where to get help

Dell support, product, and licensing information can be obtained as follows:

Product information	For product and feature documentation or release notes, go to the PowerMax Documentation page at : https://www.dell.com/support/kbdoc/en-us/000189115/powermax-info-hub-product-documentation-videos?lang=en.
Troubleshooting	For information about products, software updates, licensing, and service, go tohttps://www.dell.com/supportand locate the appropriate product support page.
Technical support	For technical support and service requests, go to https://www.dell.com/support and locate the Service Requests page. To open a service request, you must have a valid support agreement. Contact your Sales Representative for details about obtaining a valid support agreement or to answer any questions about your account.

Revision history

Document revision	Date	Description
01	July 2022	Initial release.
1.1	July 2023	Added a Note under Create NAS Server for SMB (Windows-only) file systems.

Documentation resources

Refer to the following for additional information:

Table 1. Documentation resources

Document	Description	Location
Dell EMC PowerMax File Quick Start Guide	Provides network planning and configuration information.	https://www.dell.com/support/kbdoc/en-us/000189115/powermax-info-hub-product-documentation-videos?lang=en.
Dell EMC PowerMax File Replication Guide	Provides replication planning and configuration for Synchronous or Asynchronous based File replication using PowerMax SRDF/S and SRDF/A.	
PowerMax Online Help	Provides context-sensitive information for the page opened in PowerMax Manager.	

Contents

PREFACE....3

Chapter 1: NFS Support Overview....6
NFS support....6
About secure NFS....6
Creating NFS exports....7

Chapter 2: Create NAS servers....8
Overview of configuring NAS servers with NFS protocol....8
Create a NAS server for NFS (UNIX-only) file systems....8
Configure NAS Server Naming Services....9
Configure DNS....9
Configure the NAS server UNIX Directory Service for NIS....10
Configure a NAS server UNIX Directory Service using LDAP....10
Configure NAS server to use local files for naming services....11
Configure NAS server Sharing Protocols....12
Configure NFS Server....12

Chapter 3: Configure NFS Exports....13
File systems and NFS Exports overview....13
Create a file system for NFS exports....13
Create an NFS export....14

Chapter 4: SMB Support Overview....15
SMB support....15
NAS server networks....15

Chapter 5: Create file systems and SMB shares....17
Overview of configuring NAS servers for SMB....17
Create NAS server for SMB (Windows-only) file systems....17
Change NAS server settings....18

Chapter 6: Create file systems and SMB shares....20
Create a file system....20
File system advanced settings for SMB....21
Create an SMB share....22
Advanced SMB share properties....22

Chapter 7: Global Name Space....24
Overview....24
View Global Namespace Server Links....24
Create Global Namespace....25
Create link for the Global Namespace....26
Modify Global Namespace....26
Modify Global Namespace Server Link....26

Delete Global Namespace Server....26
Delete Global Namespace Link and Target....27
Restore Namespace Server....27
Activate Namespace Server....27
GNS in Replication Environment....27

Chapter 8: Configure NAS Server Security....28
Configure FTP, or SFTP sharing protocol....28
Configure Kerberos for NAS server Security....28
Create a custom realm for Kerberos....29
Configure Kerberos security for the NAS server....30
Enable NDMP Protection and Events....30
Configure NAS server Security....31
Configure Kerberos security for the NAS server....31
Understanding Common AntiVirus Agent (CAVA)....31

Chapter 9: Additional NAS Server Features....32
Set the preferred UNIX Directory Service....32
Configure NAS server networks....32
Configure file interfaces for a NAS Server....32
Configure routes for the file interface for external connections....33
Enable NDMP Protection and Events....33

Chapter 10: More file system features....34
File system quotas....34
Enable User Quotas....35
Add a user quota onto a file system....35
Add a quota tree onto a file system....35
Add a user quota onto a quota tree....36

Index....37

1

NFS Support Overview

This chapter contains the following information:

Topics:

- NFS support
- About secure NFS
- Creating NFS exports

NFS support

PowerMax File supports NFSv3 and NFSv4. It also supports secure NFS with Kerberos, for strong authentication. While PowerMax File supports most of the NFSv4 and v4.1 functionality described in the relevant RFCs, directory delegation and pNFS are not supported. NFS support is enabled on a NAS server during or after creation, enabling you to create NFS-enabled file systems on that NAS server.

About secure NFS

You can configure secure NFS when you create or modify a NAS server that supports UNIX shares. Secure NFS provides Kerberos-based user authentication, which can provide network data integrity and network data privacy.

Kerberos is a distributed authentication service designed to provide strong authentication with secret-key cryptography. It works on the basis of "tickets" that allow nodes communicating over a non-secure network to prove their identity in a secure manner. When configured to act as a secure NFS server, the NAS server uses the RPCSEC_GSS security framework and Kerberos authentication protocol to verify users and services.

Security options

Secure NFS supports the following security options:

- krb5: Kerberos authentication
- krb5i: Kerberos authentication and data integrity by adding a signature to each NFS packet transmitted over the network
- krb5p: Kerberos authentication, data integrity, and data privacy by encrypting the data before sending it over the network

  Data encryption requires more resources for system processing and can lead to slower performance.

In a secure NFS environment, user access to NFS file systems is granted based on Kerberos principal names. However, access control to shares within a file system is based on the UNIX UID and GID, or on ACLs.

NOTE: Secure NFS supports NFS credentials with more than 16 groups, which is equivalent to the extended UNIX credentials option.

Configuring secure NFS

If you are implementing Secure NFS, configure the following:

- At least one NTP server must be configured on the PowerMax appliance to synchronize the date and time. It is recommended that you set up a minimum of two NTP servers per domain to avoid a single point of failure.
- A UNIX Directory Service (UDS)
- One or more DNS servers
- Either an AD or custom realm must be added for Kerberos authentication
- A keytab file must be uploaded to your NAS server when using a custom realm in a Kerberos configuration

Creating NFS exports

Complete the following before you can create NFS exports in PowerMax:

1. Create NAS servers with NFS protocol
2. Create a file system for NFS exports

2

Create NAS servers

This chapter contains the following information:

Topics:

- Overview of configuring NAS servers with NFS protocol
- Create a NAS server for NFS (UNIX-only) file systems
- Configure NAS Server Naming Services
- Configure NAS server Sharing Protocols

Overview of configuring NAS servers with NFS protocol

Before you can provision file storage on the storage system, a NAS server must be running on the system. A NAS server is a file server that uses the SMB protocol, NFS protocol, or both to share data with network hosts. It also catalogs, organizes, and optimizes read and write operations to the associated file systems.

This document describes how to configure a NAS server with NFS protocol, on which file systems with NFS exports can be created.

Create a NAS server for NFS (UNIX-only) file systems

You create NAS servers before creating file systems.

Prerequisites

Be sure to have your NAS network information available.

Steps

1. Select **Storage** > **File**.
2. Click on **NAS SERVERS** tab
3. Select **Create**.
4. Continue to work through the **Create NAS Server** wizard.

Wizard Screen	Description
General Details	Enter a unique NAS Server name. Select a Primary Node and Backup Node. **NOTE:** Primary and Backup node cannot be the same. Select Storage Resource Pool.
File Interface	Select Subnet Select Network Device on Primary Node and Select Network Device on Backup Node. Enter IPv4/IPv6 Address and Gateway. **NOTE:** IP address and gateway should be from the usable host IP range in the selected subnet.

Wizard Screen	Description
Protocol	Select NFS and SMB server. **NOTE:** NAS Servers can also be configured for mixed protocol support. In this case, the NAS Server exposes both NFS and SMB access for its File Systems.
DNS	Click the Enable DNS checkbox. **NOTE:** DNS server information is mandatory when: • Joining an AD domain, but optional for a stand-alone NAS server. • Configuring Secure NFS. Enter DNS Domain. Add or Delete IP address. DNS can also be used to resolve hosts defined on NFS export access lists.
UDS	You can configure the naming services with a combination of Local Files and NIS, or LDAP. Refer to the following sections for configuring: • Using Local Files • With NIS • With LDAP Select the checkbox **Enable a Unix Directory** service using NIS or LDAP. Enter NIS Domain. Add or Delete IP address.
Summary	Review the content and Select **Back** to go back and make any corrections.

5. Select **Run In The Background** to create the NAS server.
   The **Status** window opens, and you are redirected to the **NAS Servers** page once the server is listed on the page.

Next steps

Once you have created the NAS server for NFS, you can continue to configure the server settings.

If you enabled Secure NFS, you must continue to configure Kerberos.

Select the NAS server to continue to configure, or edit the **VIEW ALL DETAILS** of the NAS server settings.

Configure NAS Server Naming Services

You can configure, or modify the naming services for a NAS Server.

Naming Services include configuring one or more of the following:

- DNS
- NIS for Unix Directory Services (UDS)
- Local Files

Configure DNS

You can disable DNS, or enable and configure a NAS server to use DNS.

Prerequisites

DNS can also be used to resolve hosts defined on NFS export access lists.

DNS is required for:

- Secure NFS
- Joining an AD domain.

You cannot disable DNS for NAS servers that are configured with:

- Multiprotocol file sharing
- SMB file sharing that is joined to an Active Directory (AD)
- Secure NFS

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **NAMING SERVICES** > **DNS**.
2. Enable or disable DNS. If you enabled DNS, enter the DNS server information.

Configure the NAS server UNIX Directory Service for NIS

You can configure NAS server UNIX Directory Service (UDS) for NIS.

Steps

1. Select **Storage** > **NAS Servers** > **[nas server]** > **Naming Services** > **UDS** card.
2. If **Disabled** is on, slide the button to change to **Enabled**.
3. In the **Unix Directory Service** drop down, select **NIS**.
4. Enter an NIS **Domain** and add the IP **Addresses** for the NIS servers.
5. Select **Apply**.

Next steps

To troubleshoot issues with configuring a UDS using NIS, ensure that the NIS server domain and server IP addresses you enter are correct.

Configure a NAS server UNIX Directory Service using LDAP

About this task

You can configure a NAS server UNIX Directory Service (UDS) using LDAP.

LDAP must adhere to the IDMU, RFC2307, or RFC2307bis schemas. Some examples include AD LDAP with IDMU, iPlanet, and OpenLDAP. The LDAP server must be configured properly to provide UIDs for each user. For example, on IDMU, the administrator must go into the properties of each user and add a UID to the UNIX Attributes tab.

You can configure LDAP to use anonymous, simple, and Kerberos authentication. If using Kerberos authentication, you must configure the following before you continue to configure LDAP with Kerberos:

1. From the Naming Services card, configure the DNS server that is used to join and unjoin a Kerberos server to a realm.
2. From the Security card, add the Kerberos Realm.

Steps

1. Select **Storage** > **NAS Servers** > **[nas server]** > **Naming Services** > **UDS**
2. If **Disabled** is on, slide the button to change to **Enabled**.
3. In the **Unix Directory Service** drop down, select **LDAP**.
4. Leave the default or enter a different **Port Number**.

   NOTE: By default, LDAP uses port 389, and LDAP over SSL (LDAPS) uses port 636.

5. Add the IP addresses for the LDAP servers.

   The NAS server can be configured to use the DNS service discovery to automatically obtain LDAP server IP addresses.

   NOTE: For this discovery process to work, the DNS server must contain pointers to the LDAP servers, and the LDAP servers must share the same authentication settings.

6. Configure the LDAP authentication as described in the following table:

Wizard Screen	Description
Anonymous	Specify the Base DN, and the Profile DN for the iPlanet/OpenLDAP server.
Simple	Specify the following: ● If using AD, LDAP/IDMU: ○ Bind DN in LDAP notation format; for example, cn=administrator,cn=users,dc=svt,dc=lab,dc=com. ○ Base DN, which is the same as the Fully Qualified Domain Name (for example, svt.lab.com). ○ Profile DN. ● If using the iPlanet/OpenLDAP server: ○ Bind DN in LDAP notation format; for example, cn=administrator,cn=users,dc=svt,dc=lab,dc=com. ○ Password ○ Base DN. For example, if using svt.lab.com, the Base DN would be DC=svt,DC=lab,DC=com. ○ Profile DN for the iPlanet/OpenLDAP server.
Kerberos	Configure a custom realm to hover over any type of Kerberos realm (Windows, MIT, Heimdal). With this option, the NAS Server uses the custom Kerberos realm that is defined in the Kerberos subsection of the NAS server Security tab. **NOTE:** If you use NFS secure with a custom realm, you have to upload a keytab file.

7. Click **Apply**.

Configure NAS server to use local files for naming services

You can configure your naming services to use local files.

About this task

- Local files can be used instead of, or also with DNS, LDAP, and NIS directory services.
- If you configure local files with a UNIX Directory Service (UDS), the storage system queries the local files first.
- After you finish creating the NFS server, you can go back and upload more local files.
- Once the NAS server is created, enable the local files as described in the following steps:

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **Naming Services** > **Local Files**.
2. For each type of local file, select the down arrow to download the current file. If there is no file on the storage system, the system downloads a file template.
3. Update the file with your system information.

   To use local files for FTP access, the `passwd` file must include an encrypted password for the users. This password is used for FTP access only. The `passwd` file uses the same format and syntax as a standard UNIX system, so you can apply the password to generate the local `passwd` file. On a UNIX system, use `useradd` to add a user and `passwd` to set the password for that user. Then, copy the hashed password from the `/etc/shadow` file, add it to the second field in the `/etc/passwd` file, and upload the `/etc/passwd` file to the NAS server.
4. Save the updated file to your local machine.
5. Select **Upload Passwd Files** and browse to the location of the file you edited and select the file to upload.
6. Repeat for each type of file.

Next steps

To troubleshoot issues with configuring local files, ensure that:

- The file is created with the proper syntax. (Six colons are required for each line.) Reference the template for more details about the syntax and examples.
- Each user has a unique name and UID.

Configure NAS server Sharing Protocols

You can configure or modify the sharing protocols that are configured for a NAS server.

Configuring sharing protocols for NFS includes setting up one or more of the following:

- NFS Server
- FTP

Configure NFS Server

Configure the NAS server for UNIX-only systems, or modify the NFS server settings.

Prerequisites

DNS and NTP must be configured before configuring a Secure NFS server.

Steps

1. Select the **Storage** > **File** > **NAS Servers** > **[nas server]** > **Sharing Protocols** > **NFS Server** tab.
2. If **Disabled** is on, slide the button to change to **Enabled** to define the NAS server.
3. Enable either **NFSv3**, **NFSv4**, or both.
4. Optionally, disable, or enable Secure NFS.
   Extended UNIX credentials are also enabled.
5. **Enable or disable Extend Unix** credentials.

   NOTE: Secure NFS supports NFS credentials with more than 16 groups, which is equivalent to the extended UNIX credentials option.

   - If this field is selected, the NAS server uses the User ID (UID) to obtain the primary Group ID (GID) and all group GIDs to which it belongs. The NAS server obtains the GIDs from the local password file or UDS.
   - If this field is cleared, the UNIX credential of the NFS request is directly unzipped from the network information that is contained in the frame. This method has better performance, but it is limited to including up to only 16 group GIDs.
6. In the **Credential Cache Retention**, enter a time period (in minutes) for which access credentials are retained in the cache.
7. **Apply** the changes.

3

Configure NFS Exports

This chapter contains the following information:

Topics:

- File systems and NFS Exports overview
- Create a file system for NFS exports
- Create an NFS export

File systems and NFS Exports overview

While creating File Systems and NFS Exports, it is helpful to note the following:

- A NAS server must be configured to support NFS protocol before creating a file system.
- You can choose to add NFS Exports the first time you create the file system, or you can add NFS Exports to a file system after it has been created.

Create a file system for NFS exports

You can create a file system for NFS exports.

Prerequisites

Make sure that there is a NAS server that is configured to support the NFS protocol.

Steps

1. Select **Storage** > **File Systems**.
2. Click **Create**.
   The **Create File System** wizard launches.

Option	Description
Select Type	**Select Type** Select the file system you want to create: General (Select this if you want to create a general NAS File System) Or VMware File System (Select this if you want to create NFS Datastore. VMware File System can only be accessed using NFS).
Select NAS Server	Select NAS Server with NFS Server Enabled and/or SMB Server Enabled
File System Details	Enter Name, Description, Size, Service Level. Set Thresholds.
FLR	Select File Level Retention. FLR will protect files from modification and deletion during the retention period.

Option	Description
	NOTE: Once File System is created, FLR cannot be enabled, disabled, or change to another type.
NFS Export	Optionally enable the Configure the Initial Export
Summary	Review the content and Select **Back** to go back and make any corrections.

3. Review the summary and click **Run In The Background**.
   The file system is added to the **File System** tab. If you created an export simultaneously, the export displays in the **NFS export** tab.

Create an NFS export

You can create an NFS export on a file system.

Steps

1. Select the **Storage** > **File** > **File Systems** > **NFS Export** tab.
2. Click **Create**.
   The **Create NFS Export** wizard launches.
3. Enter the requested information while noting the following:
   - Snapshots must have been created before creating the NFS export.
   - **Local Path** must correspond to an existing folder name within the file system that was created from the host-side.
   - The value specified in the **NFS Export Details**, **Name** field, along with the NAS server name, constitutes the name by which hosts access the export.
   - NFS export names must be unique at the NAS server level per protocol. However, you can specify the same name for an SMB share, and NFS exports.
4. Once you approve the settings, click **Create NFS Export**.
   The NFS Export displays on the **NFS Export** page.

4

SMB Support Overview

This chapter contains the following information:

Topics:

- SMB support
- NAS server networks

SMB support

PowerMax File supports SMB 1 through SMB 3.1.1. When SMB support is enabled on the NAS server, you can create SMB-enabled file systems. The NAS server with SMB support can either be stand-alone or Active Directory domain-joined. Domain-joined NAS servers are placed in the OU=Computers, OU=EMC NAS Servers organizational unit by default.

SMB file systems and shares have the following advanced protocol options:

NOTE: These options, except for Oplocks Enabled, are disabled by default.

Table 2. SMB advanced protocol options

Protocol option	Level
Sync Writes Enabled	File system
Oplocks Enabled	File system
Notify on Write Enabled	File system
Notify on Access Enabled	File system
Continuous Availability	Share
Protocol Encryption	Share
Access-Based Enumeration	Share
Branch Cache Enabled	Share
Offline Availability	Share

NAS server networks

Configure the following before configuring NAS servers with SMB protocol:

1. Configure one or more DNS servers.
2. If you are joining the NAS server to the Active Directory (AD), configure at least one NTP server on the storage system to synchronize the date and time. It is recommended that you set up a minimum of two NTP servers per domain to avoid a single point of failure.

   NOTE: During AD creation, NTP is configured.
3. Create a domain account in Active Directory.

Creating SMB shares

Complete the following before you can create SMB shares in PowerMax:

1. Create NAS servers with SMB protocol

2. Create a file system for SMB shares

5

Create file systems and SMB shares

This chapter contains the following information:

Topics:

- Overview of configuring NAS servers for SMB
- Create NAS server for SMB (Windows-only) file systems
- Change NAS server settings

Overview of configuring NAS servers for SMB

This document describes how to configure a NAS server with SMB protocol, on which file systems with SMB shares can be created.

Create NAS server for SMB (Windows-only) file systems

You create a NAS server before creating file systems.

Prerequisites

Obtain the following information:

- Network port, IP Address, Subnet Mask/Prefix Length, Gateway information for the NAS Server.
  NOTE: IP Address and Subnet Mask/Prefix Length are mandatory.
- If you are configuring a stand-alone NAS server, obtain the workgroup and NetBIOS name. Then define what to use for the stand-alone local administrator of the SMB server account.
- If you are joining the NAS server to the Active Directory (AD), ensure that NTP is configured on your storage system. Then obtain the SMB computer name (used to access SMB shares), Windows domain name, and the username and password of a domain administrator or domain user who has a sufficient domain access level to join the AD.

Steps

1. Select **Storage** > **File** > **NAS Servers**.
2. Select **Create**.
3. Continue to work through the **Create NAS Server** wizard.

Wizard Screen	Description
Details	Enter a NAS server name, description, and network details.
Sharing Protocol	**Select Sharing Protocol** Select **SMB**. **NOTE:** If you select both SMB and NFS protocols, you automatically enable the NAS server to support multiprotocol. Multiprotocol configuration is not described in this document. **Windows Server Settings** Select **Standalone** to create a stand-alone SMB server or **Join to the Active Directory Domain** to create a domain member SMB server.

Wizard Screen	Description
	If you join the NAS server to the AD, optionally Select **Advanced** to change the default NetBios name and organizational unit. **DNS** If you selected to **Join to the Active Directory Domain**, it is mandatory to add a DNS server. Optionally, enable DNS if you want to use a DNS server for your stand-alone SMB server. **User Mapping** The **User Mapping** page displays if you have selected to join the active directory domain. Keep the default **Enable automatic mapping for unmapped Windows accounts/users**, to support joining the active directory domain. Automatic mapping is required when joining the active directory domain.
Summary	Review the content and Select **Previous** to go back and make any corrections.

4. Select **Create NAS Server**.
   The **Status** window opens, and you are redirected to the **NAS Servers** page once the server is created.

Next steps

Once you have created the NAS server for SMB, you can continue to configure the server settings, or create file systems.

Select the NAS server to continue to configure, or modify the NAS server settings.

NOTE: As part of SMB configuration for NAS Servers, it is recommended to set 'Continuous Availability (CA)' property to ensure transparent failover of Filesystem access by Clients.

Change NAS server settings

Once you have created a NAS server, you can make configuration changes to the server.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]**.
2. On the **Network** page, optionally configure the network interfaces or the routes to external networks as described in Configure NAS server networks.
3. On the **Naming Services** page, optionally add, modify, or delete NAS server DNS servers.

   NOTE: You cannot disable DNS for NAS servers that support SMB file sharing and that are joined to an Active Directory (AD).

4. On the **Sharing Protocols** page:
   - Select the **SMB Server** card to enable or disable support for Windows shares, or to change the type of lookup the SMB server will use.

     NOTE: If you change the **Windows Server Type** from **Standalone** to **Join to the Active Directory Domain**, then you must go to the **User Mapping** tab and select **Enable automatic mapping for unmapped Windows accounts/users**.
   - Select the **FTP** card to enable or disable FTP or SFTP, change FTP or SFTP properties, configure user authentication, a user home directory, and authentication message settings.

     For details see Configure FTP Sharing Protocol.
   - Select **User Mapping** to enable the server to use automatic mapping for unmapped Windows account/users, or the default account for unmapped Windows account users.
5. On the **Protection & Events** page, enable or disable NDMP.

   For details see Enable NDMP Protection and Events.
6. On the **Security** tab:

- Select **Kerberos** to add the active directory (AD) realm for Kerberos authentication or to configure a custom Kerberos realm.
- Select **Antivirus** to enable or disable the anti-virus service and to retrieve or upload the anti-virus configuration file.

For details see Configure NAS Server Security

6

Create file systems and SMB shares

This chapter contains the following information:

Topics:

- Create a file system
- Create an SMB share

Create a file system

A file system must be created on the NAS server before you can create an SMB share.

Prerequisites

Make sure that there is a NAS server that is configured to support the SMB protocol as described in Configuring NAS servers.

Steps

1. Select **Storage** > **File** > **File Systems** and click **Create**.
2. Continue to work through the **Create File System** wizard.

Option	Description
Select NAS Server	Select a NAS server enabled for SMB.
Advanced SMB Settings	Optionally choose from the following: • **Sync Writes Enabled** • **Oplocks Enabled** • **Notify on Write Enabled** • **Notify on Access Enabled** • **Enable SMB Events Publishing** For details see File system advanced settings for SMB shares.
File System Details	Provide the file system name, and the size of the file system. The file system size can be from 3 GB to 256 TB. **NOTE:** All thin file systems, regardless of size, have 1.5 GB reserved for metadata upon creation. For example, after creating a 100GB thin file system, PowerMax T model immediately shows 1.5 GB used. When the file system is mounted to a host, it shows 98.5 GB of usable capacity. This is because the metadata space is reserved from the usable file system capacity.
SMB Share	Optionally, configure the initial SMB Share. You can add shares to the file system after the initial file system configuration. For details about the SMB Share options, see: Create an SMB share.
Protection Policy	Optionally, provide a protection policy for the file system. **NOTE:** PowerMax supports snapshots for file systems. If a protection policy is set for both replication and snapshot protections, PowerMax implements the snapshot policy on the file system, and ignore the replication policy for the file system.
Summary	Review the summary. Go back to make necessary updates.

3. Click **Create File System**.
   The file system is displayed in the File System list, and if you created an SMB Share, it is displayed in the SMB Share list.

File system advanced settings for SMB

You can add advanced settings to SMB-enabled file systems while creating a file system.

Table 3. File system advanced settings for SMB

Setting	Description
Sync Writes Enabled	When you enable the synchronous writes option for a Windows (SMB) or multiprotocol file system, the storage system performs immediate synchronous writes for storage operations, regardless of how the SMB protocol performs write operations. Enabling synchronous writes operations enables you to store and access database files (for example, MySQL) on storage system SMB shares. This option guarantees that any write to the share is done synchronously and reduces the chances of data loss or file corruption in various failure scenarios, for example, loss of power. This option is disabled by default. **NOTE:** The synchronous writes option can have a significant impact on performance. It is not recommended unless you intend to use Windows file systems to provide storage for database applications.
Oplocks Enabled	(Enabled by default) Opportunistic file locks (oplocks, also known as Level 1 opslock) enable SMB clients to buffer file data locally before sending it to a server. SMB clients can then work with files locally and periodically communicate changes to the storage system rather than having to communicate every operation over the network to the storage system. This feature is enabled by default for Windows (SMB) and multiprotocol file systems. Unless your application handles critical data or has specific requirements that make this mode or operation unfeasible, leaving the oplocks enabled is recommended. The following oplocks implementations are supported: • Level II oplocks, which informs a client that multiple clients are accessing a file, but no client has yet modified it. A level II oplock lets the client perform read operations and file attribute fetches by using cached or read-ahead local information. All other file access requests must be sent to the server. • Exclusive oplocks, which informs a client that it is the only client opening the file. An exclusive oplock lets a client perform all file operations by using cached or read-ahead information until it closes the file, at which time the server must be updated with any changes that are made to the state of the file (contents and attributes). • Batch oplocks, which informs a client that it is the only client opening the file. A batch oplock lets a client perform all file operations by using cached or read-ahead information (including opens and closes). The server can keep a file opened for a client even though the local process on the client machine has closed the file. This mechanism curtails the amount of network traffic by letting clients skip the extraneous close and open requests.
Notify on Write Enabled	Enable notification when a file system is written to. This option is disabled by default.
Notify on Access Enabled	Enable notification when a file system is accessed. This option is disabled by default.
Enable SMB Events publishing	Enable the processing of SMB events for this file system.

Create an SMB share

You can create an SMB share on a file system that has been created with an SMB-enabled NAS server.

Steps

1. Select **Storage** > **File** > **File System** > **SMB Share**.
2. Click **Create** and continue to work through the **Create SMB Share** wizard.

Option	Description
Select File System	Select a file system that has been enabled for SMB.
Select a snapshot of the file system	Optionally, select one of the file system snapshots on which to create the share. Only snapshots are supported for file system protection policies. Replication is not supported for file systems.
SMB Share Details	Enter a name, and local path for the share. When entering the local path: • You can create multiple shares with the same local path on a single SMB file system. In these cases, you can specify different host-side access controls for different users, but the shares within the file system have access to common content. • A directory must exist before you can create shares on it. If you want the SMB shares within the same file system to access different content, you must first create a directory on the Windows host that is mapped to the file system. Then, you can create corresponding shares using PowerMax. You can also create and manage SMB shares from the Microsoft Management Console. PowerMax also created the SMB Share path, which uses the host to connect to the share. The export path is the IP address of the file system, and the name of the share. Hosts use either the file name or the share path to mount or map to the share from a network host.
Advanced SMB Properties	Enable one or more of the Advanced SMB Settings. • Continuous Availability • Protocol Encryption • Access-based Enumeration • Branch Cache Enabled Decide which objects are available when the share is offline. For details see Advanced SMB properties.
Summary	Review the summary. Go back to make necessary updates.

Next steps

Once you create a share, you can modify the share from PowerMax or using the Microsoft Management Console.

To modify the share from PowerMax, select the share from the list on the **SMB Share** page, and click **Modify**.

Advanced SMB share properties

You can configure the following advanced SMB share properties when you create an SMB share or change its properties:

Table 4. Advanced SMB Properties

Option	Description
Continuous Availability	Gives host applications transparent, continuous access to a share following a failover of the NAS server on the system (with the NAS server internal state saved or restored during the failover process). **NOTE:** Enable continuous availability for a share only when you want to use Microsoft Server Message Block (SMB) 3.0 protocol clients with the specific share.

Table 4. Advanced SMB Properties (continued)

Option	Description
Protocol Encryption	Enables SMB encryption of the network traffic through the share. SMB encryption is supported by SMB 3.0 clients and above. By default, access is denied if an SMB 2 client attempts to access a share with protocol encryption enabled. You can control this by configuring the RejectUnencryptedAccess registry key on the NAS Server. 1 (default) rejects non-encrypted access and 0 allows clients that do not support encryption to access the file system without encryption.
Access-Based Enumeration	Filters the list of available files and directories on the share to include only those to which the requesting user has read access. **NOTE:** Administrators can always list all files.
Branch Cache Enabled	Copies content from the share and caches it at branch offices. This allows client computers at branch offices to access the content locally rather than over the WAN. BranchCache is managed from Microsoft hosts.
Offline Availability	Configures the client-side caching of offline files: • **Manual**: Files are cached and available offline only when caching is explicitly requested. • **Programs and files opened by users**: All files that clients open from the share are automatically cached and available offline. Clients open these files from the share when they are connected to it. This option is recommended for files with shared work. • **Programs and files opened by users, optimize for performance**: All files that clients open from the share are automatically cached and available offline. Clients open these files from the share's local cache, if possible, even when they are connected to the network. This option is recommended for executable programs. • **None**: Client-side caching of offline files is not configured.

7

Global Name Space

This chapter contains the following information:

Topics:

- Overview
- View Global Namespace Server Links
- Create Global Namespace
- Create link for the Global Namespace
- Modify Global Namespace
- Modify Global Namespace Server Link
- Delete Global Namespace Server
- Delete Global Namespace Link and Target
- Restore Namespace Server
- Activate Namespace Server
- GNS in Replication Environment

Overview

GNS (Global Namespace) provides namespaces in a similar way as Microsoft DFS management tool. PowerMax File GNS enables this functionality by grouping shared folders (or exports) located on different servers into one or more logically structured namespaces. It can access file storage with a single share/export (single access point), presenting all FSs a directory structure. It gives users a virtual view of shared folders, where a single path leads to files located on multiple smb/nfs servers. PowerMax File provides this ability to create and manage GNS. The system provides the ability to create GNS Links and then user can add a targeted share/export to GNS. Multiple Links can be added to the same GNS and each Link may reside in a nested path relative to the GNS root folder (e.g.: GNS_ROOT/a/b/c/Link→some_target). The target folder might reside other than GNS hosting FS inside the same NAS server as GNS root and FS on the NAS server, which is other than the one that hosts SNS.

Prerequisites

1. SMB Server must be configured properly for the relevant NAS Server for GNS of SMB type to be created.
2. NFS Server (nfsv4) must be configured properly for the relevant NAS Server for GNS of NFS type to be created.
3. GNS name should be unique across the same NAS Server and must not conflict with share/export name on the same NAS Server. This limitation is due to a fact that GNS internally implemented via DFS enabled SMB Share.

View Global Namespace Server Links

About this task

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. The list of snapshots can be viewed with the following details:

   NAS Server Name

   · Filesystem Name

   · Health

- Number of Links
- Number of Alerts

3. Select a **Namespace Server Name** to view the following details:
   - Path used by remote hosts to connect to the target FS.
   - UNC path - a compound of remote Server FQDN or IP address, share/export name, and path.

Create Global Namespace

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Click **Create** and continue to work through the **Global Namespace Server** wizard.

Option	Description
Select Type	Select Type of Global Namespace Server(GNS) you want to create: ● Namespace server for NFS only - The namespace provides access via NFS(nfsv4) protocol only. ● Namespace server for SMB only - The namespace provides access via SMB protocol only. ● Namespace server for NFS and SMB - The namespace provides access via both NFS and SMB protocols. Click **Next**.
Select NAS Server	Select NAS Server. NAS Server can host multiple Namespaces. Click **Next**.
Select FileSystem	Select FileSystem Option 1: Create its own File System (Recommended) Create its own Filesystem to host GNS is recommended so that File System does not have exports/ shares defined as it is impossible to create an NFS export pointing to a root of this FS for the existing created FS. Enter Name, description, size, storage resource pool and select the service level. Click **Next**. Or Option 2: Select from available General Type File system User must take care of selecting FS not having its NFS export pointing to a root of this FS, else it will result into failure of creation of GNS server. Select filesystems having exports not mounted on root directory and click **Next**. **NOTE:** Namespace filesystem can host many Namespace Servers if the paths are separate and not conflicting.
Namespace Details	Provide Namespace Details Name of the server ● It is used by remote hosts to connect to the Unified Namespace over the network Local Path ● Local path to the storage resource or any existing sub-folder of the storage resource that is shared over the network. ● This path is relative to the NAS Server and thus must start with the filesystem's mountpoint path, which is the filesystem name. ● For example, to share the top-level of a filesystem named svr1fs1, which is mounted on the /svr1fs1 mountpoint on the NAS Server, use /svr1fs1 in the path parameter.

Option	Description
	Namespace Path will be the formed based on the namespace server names and interface of the selected NAS Server. Click **Next**.

3. The Global Namespace is displayed in the Namespace server list.

Create link for the Global Namespace

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Select a Global Namespace Server from the list view.
3. Click **Create** and continue to work through the **Create Link for the Namespace** wizard.
4. Enter **Local Path**.

   NOTE: A path name relative to the namespace root (without a forward slash or trailing slash), that is used by remote hosts to connect to the target FS.

5. Enter Description and Client Cache Timeout.
6. Add Target UNC Path (a compound of remote Server FQDN or IP address, share/export name, and path).
7. Click **OK**.

Modify Global Namespace

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Select the check box against the Global Namespace Server from the list view.
3. Click **Modify** and continue to work through the **Modify Global Namespace Server** wizard.
4. Enter Description, Type of Namespace and Client Cache Timeout.
5. Click **OK**.

Modify Global Namespace Server Link

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Select a Global Namespace Server from the list view.
3. Click **Modify** and continue to work through the **Modify Link and Target of the Namespace** wizard.
4. Enter Description and Client Cache Timeout.
5. Optionally modify the referral status of link target.
6. Click **OK**.

Delete Global Namespace Server

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.

2. Select the check box against the Global Namespace Server from the list view.
3. Click the **Delete** icon. The **Delete Global Namespace Server** dialogue box opens.
4. Click **OK**.

Delete Global Namespace Link and Target

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Select a Global Namespace Server from the list view.
3. Click the **Delete** icon. The **Delete Link for the Namespace** dialogue box opens.
4. Click **OK**.

Restore Namespace Server

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Select the check box against the Global Namespace Server from the list view.
3. Click **Restore**.

Activate Namespace Server

Steps

1. Select **Storage** > **File** >**GLOBAL NAMESPACES** tab.
2. Select the check box against the Global Namespace Server from the list view.
3. Click **Activate**.

GNS in Replication Environment

This Nameserver (hosted on selected Namespace server) and its links/targets will be available on destination site after failover/switchover on the new primary site. When NAS Server are replicated after the failover/switchover, admins should take care of FS hosting GNS to be visible after the operation and make sure that the links are active after failover/switchover on the FS hosted GNS.

8

Configure NAS Server Security

This chapter contains the following information:

Topics:

- Configure FTP, or SFTP sharing protocol
- Configure Kerberos for NAS server Security
- Enable NDMP Protection and Events
- Configure NAS server Security

Configure FTP, or SFTP sharing protocol

You can configure FTP or FTP over SSH (SFTP) settings for an existing NAS server only.

Prerequisites

Passive mode FTP is not supported.

FTP access can be authenticated using the same methods as NFS. Once authentication is complete, access is the same as NFS for security and permission purposes. If the format is anything other than `user@domain` or `domain\user`, NFS authentication is used. NFS authentication uses local files, LDAP, NIS, or local files with LDAP or NIS.

To use local files for NFS, FTP access, the `passwd` file must include an encrypted password for the users. This password is used for FTP access only. The `passwd` file uses the same format and syntax as a standard Unix system, so you can leverage this to generate the local `passwd` file. On a Unix system, use `useradd` to add a new user and `passwd` to set the password for that user. Then, copy the hashed password from the `/etc/shadow` file, add it to the second field in the `/etc/passwd` file, and upload the `/etc/passwd` file to the NAS server.

Steps

1. Select the **Storage** > **File** > **NAS Servers** > **[nas server]** > **Sharing Protocols** > **FTP** tab.
2. Under **FTP**, if Disabled in on, slide the button to **Enable**.
3. Optionally also enable SSH FTP. Under **SFTP**, if Disabled in on, slide the button to **Enable**.
4. **Under FTP/SFTP Server Access**, Select which type of authenticated users have access to the files.
5. Optionally, show the **Home Directory and Audit** options.
   - Select or clear the **Home directory restrictions**. If disabled, enter the **Default home directory**.
   - Select or clear **Enable FTP/SFTP Auditing**. If checked, enter the directory location of where to save the audit files, and the maximum size allowed for the audit file.
6. Optionally, **Show Messages**, and enter a default **Welcome message**, and **Message of the day**.
7. Optionally, **Show Access Control List** to provide access or deny access to **Filtered Users**, **Filtered Groups**, and **Filtered hosts**.
8. Click **Apply**.

Configure Kerberos for NAS server Security

You can configure the NAS Server with Kerberos.

Kerberos is a distributed authentication service designed to provide strong authentication with secret-key cryptography. It works on the basis of "tickets" that allow nodes communicating over a non-secure network to prove their identity in a secure manner. When configured to act as a secure NFS server, the NAS server uses the RPCSEC_GSS security framework and Kerberos authentication protocol to verify users and services.

If the NAS server has been configured with NFS only, and you are configuring Secure NFS, or LDAP with Kerberos, you must configure Kerberos with a custom realm before configuring security in PowerMax.

If the NAS server has been configured with both the NFS and SMB protocol, you have the option of using Kerberos that is inherited with AD since the domain joined SMB server exists on the NAS server.

The storage system must be configured with an NTP server. Kerberos relies on the correct time synchronization between the KDC, servers, and client on the network.

Configuring Kerberos for Secure NFS

If you are configuring Kerberos for Secure NFS, be aware of the following:

- If configuring the NAS server for NFS only, you must configure the NAS server with a custom realm. If you have configured the NAS server with NFS and SMB, you can use either the AD or custom realm.
- Using LDAPS or LDAP with Kerberos is recommended for increased security.
- A DNS server must be configured at the NAS-server level. All members of the Kerberos realm, including the KDC, NFS server, and NFS clients, must be registered in the DNS server.
- The NFS client's hostname FQDN and NAS server FQDN must be registered in the DNS server. Clients and servers must be able to resolve any member of the Kerberos realm's FQDNs to an IP address.
- The FQDN part of the NFS client's SPN must be registered in the DNS server.
- A keytab file must be uploaded to your NAS server when configuring Secure NFS.

Create a custom realm for Kerberos

You can configure a custom realm to use with Kerberos.

A custom Kerberos realm lets you configure any kind of KDC (MIT/Heidmal or AD). Use this method when you do not have an SMB server domain that is configured on the NAS server or if you want to use a different Kerberos realm than the one configured for the SMB server.

Create custom realm for pure NFS Server

To use a Unix-based KDC, follow these steps before configuring Kerberos in PowerMax. The steps assume that you want to use myrealm in the Kerberos realm linux.dellemc.com as the hostname of the NFS server.

1. Run the `kadmin.local` tool.
2. Create the principals and their keys:

   ```
   kadmin.local: addprinc -randkey nfs/myrealm.linux.dellemc.com
   ```

   and/or

   ```
   kadmin.local: addprinc -randkey nfs/myrealm
   ```

3. Put the key of the principal into the keytab file myrealm.linux.dellemc.fr:

   ```
   kadmin.local: ktadd -k myrealm.linux.dellemc.com.keytab nfs/myrealm.linux.dellemc.fr
   ```

Create custom realm for multiprotocol (NFS and SMB) NAS server

To use a Windows-based KDC without using the SMB server account on the NAS server, follow these steps before configuring Kerberos in PowerMax. The steps assume that you want to use myrealm.windows.dellemc.com as the FQDN for the NFS server.

1. Create account myrealm for the NAS server in the Active Directory (AD) of the windows domain windows.dellemc.com.
2. Register the service SPN on the computer account you created:

   ```
   C:\setspn -S nfs/myrealm.windows.dellemc.com myrealm
   ```

3. Verify that the SPN was created.

```
C:\setspn myrealm
```

4. Generate a keytab file for the SPN:

```
C:\ktpass -princ nfs/myrealm.windows.dellemc.com@WINDOWS.DELLEMC.COM -mapuser
WINDOWS\myrealm
 -crypto ALL +rndpass -ptype KRB5_NT_PRINCIPAL -out myrealm.windows.dellemc.com.keytab
```

Configure Kerberos security for the NAS server

You can configure the NAS server with Kerberos security.

About this task

If configuring for NFS, DNS and UDS must be configured for the NAS server and all members of the Kerberos realm must be registered in the DNS server.

If using a NAS server that is configured for both SMB and NFS, be sure to add the SMB server to the AD domain.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **Security** > **Kerberos**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Enter the name of the **Realm**.
4. Enter the **Kerberos IP Address** and click **Add**.
5. Enter the TCP Port for Kerberos to use. 88 is the default port.
6. Click **Apply**.

Next steps

If you choose to change from an AD realm to a custom realm after the NAS server is successfully created with Secure NFS, you cannot mount any NFS exports until you perform the following operations:

1. Create a Keytab file.
2. Remove the AD realm from the NAS server.
3. Enter the Username and Password for the AD Server.
4. Enter the custom realm.
5. Upload the Keytab file.

Enable NDMP Protection and Events

You can configure standard backup for the NAS servers using NDMP. The Network Data Management Protocol (NDMP) provides a standard for backing up file servers on a network. Once NDMP is enabled, a third-party Data Management Application (DMA), such as Dell EMC Networker, can detect the PowerMax NDMP using the NAS server IP address.

About this task

Enabling NDMP is performed after the NAS server is created.

PowerMax supports:

- Three-way NDMP. The data is transferred through the DMA over a local area network (LAN) or Wide Area Network (WAN).
- Full and incremental backups.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **Protection and Events**.
2. Under **BACKUP & EVENTS**, if **Disabled** is on, slide the button to change to **Enabled**.
3. Enter a password for the **New Password**.

The user name is always `ndmp`.

4. Re-enter the same password as the new password in **Verify Password**.
5. Click **Apply**.

Next steps

Leave the NDMP page, and return back to the NDMP page to validate that NDMP is enabled.

Configure NAS server Security

You can configure the NAS server with **Kerberos** or **Antivirus** security.

Configuring NAS server security includes the following options:

- Kerberos
- Antivirus

Configure Kerberos security for the NAS server

You can configure the NAS server with Kerberos security.

About this task

Be sure to add the SMB server to the AD domain before configuring Kerberos.

If you are configuring the NAS server for SMB-only, you do not need a Keytab file. Keytab file is only required for Secure NFS configuration.

Steps

1. Select **Storage** > **NAS Servers** > **[nas server]** > **Security** > **Kerberos**.
2. If Disabled is on, slide the button to change to **Enabled**.
3. Enter the name of the **Realm**.
4. Enter the Kerberos IP Address and click **Add**.
5. Enter the TCP Port to use for Kerberos. 88 is the default port.
6. Click **Apply**.

Understanding Common AntiVirus Agent (CAVA)

Common AntiVirus Agent (CAVA) provides an antivirus solution to clients using a NAS server. It uses an industry-standard SMB protocol in a Microsoft Windows Server environment. CAVA uses third-party antivirus software to identify and eliminate known viruses before they infect files on the storage system.

Antivirus software is important because the storage system is resistant to the invasion of viruses because of its architecture. The NAS server runs data access in real-time using an embedded operating system. Third parties are unable to run programs containing viruses on this operating system. Although the operating system software is resistant to viruses, Windows clients that access the storage system require virus protection. Virus protection on clients reduces the chance that they will store an infected file on the server, and protects them if they open an infected file. This antivirus solution consists of a combination of the operating system software, CAVA agent, and a third-party antivirus engine. The CAVA software and a third-party antivirus engine must be installed on a Windows Server in the domain.

For the CEE CAVA versions required by see the . For additional information about CAVA, which is part of Common Event Enabler (CEE), see *Using the Common Event Enabler on Windows Platforms* on https://www.dell.com/support.

9

Additional NAS Server Features

This chapter contains the following information:

Topics:

- Set the preferred UNIX Directory Service
- Configure NAS server networks
- Enable NDMP Protection and Events

Set the preferred UNIX Directory Service

After you have created a NAS server, you can set the preferred UNIX Directory Services (UDS) search order for user access.

Steps

1. Select **Storage** > **File** > **NAS Servers**.
2. Select the checkbox in the **Name** column to the left of the NAS server.
3. Click **Modify**.
4. Select the preferred UDS search order for use from the list of **Unix Directory Service Search Order** drop down.
5. Click **Apply**.

Configure NAS server networks

You can modify or configure NAS server networks.

Configure the following for NAS server networks:

- The file interfaces
- Routes to external services such as hosts

Configure file interfaces for a NAS Server

You can configure the file interfaces for a NAS server after the server has been added to PowerMax.

About this task

You can add more file interfaces, and define which is the preferred interface to use. Also, you can define which interface to use for production and backup, or for IPv4, or IPv6.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]**.
2. Click **Add** to add another file interface to the NAS server.
3. Enter the File Interface properties.

   NOTE: You cannot reuse VLANs that are being used for the management and storage networks.

4. You can perform the following on a File Interface by selecting a file interface from the list. Click:

Option	Description
Modify	To change the properties of the file interface properties.

Option	Description
Delete	To delete the file interface from the NAS server.
Ping	To test the connectivity from the NAS server to the external IP address.
Preferred Interface	To define which interface PowerMax should default to using when multiple production and backup interfaces have been defined.

Configure routes for the file interface for external connections

You can configure the routes that the file system uses for external connections.

Prerequisites

You can use the **Ping** option from the **File Interface** card to determine if the file interface has access to the external resource.

About this task

Usually, the NAS server interfaces are configured with a default gateway, which is used to route requests from the NAS server interface to external services.

Use the following steps:

- If you need to configure more granular routes to external services.
- To add a route to access a server from a specific interface through a specific gateway.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **Network** > **Routes to External Services**.
2. Click **Add** to enter the route information in the **Add Route** wizard.

Enable NDMP Protection and Events

You can configure standard backup for the NAS servers using NDMP. The Network Data Management Protocol (NDMP) provides a standard for backing up file servers on a network. Once NDMP is enabled, a third-party Data Management Application (DMA), such as Dell EMC Networker, can detect the PowerMax NDMP using the NAS server IP address.

About this task

Enabling NDMP is performed after the NAS server is created.

PowerMax supports:

- Three-way NDMP. The data is transferred through the DMA over a local area network (LAN) or Wide Area Network (WAN).
- Full and incremental backups.

Steps

1. Select **Storage** > **File** > **NAS Servers** > **[nas server]** > **Protection** .
2. Under **NDMP Backup**, if **Disabled** is on, slide the button to change to **Enabled**.
3. Enter a password for the **New Password**.
   The user name is always `ndmp`.
4. Re-enter the same password as the new password in **Verify Password**.
5. Click **Apply**.

Next steps

Leave the NDMP page, and return back to the NDMP page to validate that NDMP is enabled.

10

More file system features

This chapter contains the following information:

Topics:

- File system quotas

File system quotas

You can track and limit drive space consumption by configuring quotas for file systems at the file system or directory level. You can enable or disable quotas at any time, but it is recommended that you enable or disable them during non-peak production hours to avoid impacting file system operations.

NOTE: You cannot enable quotas for read-only file systems.

Types of quotas

There are three types of quotas you can put on a file system.

Table 5. Quota types

Type	Description
User Quotas	Limits the amount of storage that is consumed by an individual user storing data on the file system.
Tree Quota	Tree quotas limit the total amount of storage that is consumed on a specific directory tree. You can use tree quotas to: • Set storage limits on a project basis. For example, you can establish tree quotas for a project directory that has multiple users sharing and creating files in it. • Track directory usage by setting the tree quota hard and soft limits to 0 (zero). **NOTE:** If you change the limits for a tree quota, the changes take effect immediately without disrupting file system operations.
User quota on a quota tree	Limits the amount of storage that is consumed by an individual user storing data on the quota tree.

Quota Limits

Table 6. Hard and Soft Limits

Type	Descriptions
Hard	A hard limit is an absolute limit on storage usage. If a hard limit is reached for a user quota on a file system or quota tree, the user cannot write data to the file system or tree until more space becomes available. If a hard limit is reached for a quota tree, no user can write data to the tree until more space becomes available.
Soft limit	A soft limit is a preferred limit on storage usage. The user is allowed to use space until a grace period has been reached. The user is alerted when the soft limit is reached, until the grace period is over. After that, an out of space condition is reached until the user gets back under the soft limit.

Quota Grace Period

The Quota Grace Period, provides the ability to set a specific grace period to each tree quota on a file system. The grace period counts down the time between the soft and hard limit, and alerts the user about the time remaining before the hard limit is met. If the grace period expires you can not write to the file system until more space has been added, even if the hard limit has not been met.

You can set an expiration date for the Grace Period. The default is 7 days, alternatively you can set the Grace Period expiration date to an infinite amount of time and the Grace Period will never expire, or for specified number of days, hours or minutes. Once the Grace Period expiration date is met, the Grace Period will no longer apply to the File System directory.

Additional information

For more information on quotas, see the *Dell EMC PowerMax File Capabilities White Paper*.

Enable User Quotas

You must enable Quotas and set the User Quota defaults before you can add a User Quota to a files system.

Steps

1. Select **Storage** > **File** > **File Systems** > **[file system]** > **Quotas**.
2. Select **Storage** > **File** > **File Systems** > **[file system]** > **Quotas** > **Properties**.
3. Slide the **Disabled** button to the right until it is **Enabled**.
4. Enter the default **Grace Period** for the user quota on the file system which will count down time after the soft limit is met until the hard limit will be met.
5. Enter a default **Soft Limit**, and a default **Hard Limit** and click **Update**.

Add a user quota onto a file system

Create a user quota on a file system to limit or track the amount of storage space that individual users consume on that file system. When you create or modify user quotas, you can use default hard and soft limits that are set at the file-system level.

Prerequisites

You must enable Quotas and set the User Quota defaults before you can add a User Quota to a files system. See Enable User Quotas.

NOTE: You cannot create quotas for read-only file systems.

Steps

1. Select **Storage** > **File** > **File Systems** > **[file system]** > **Quotas** > **User**.
2. Select **Add** on the **User Quota** page.
3. In the **Add User Quota** wizard, provide the requested information. To track space consumption without setting limits, set **Soft Limit** and **Hard Limit** to 0, which indicates no limit.
4. Select **Add**.

Add a quota tree onto a file system

About this task

Create a quota tree at the directory level of a file system to limit or track the total storage space that is consumed for that directory.

Steps

1. Select **Storage** > **File** > **File Systems** > **[file system]** > **Quotas** > **Tree Quotas**.
2. Select **Add**.
3. Slide the **Enforce User Quota** to the right to enabled User Quota defaults on the Tree Quota.
4. Provide the requested information.
   - Enter a **Grace Period** to count down the time between the soft and hard limit. You will begin to receive alerts once the grace period is reached.
   - To track space consumption without setting limits, set the **Soft Limit** and **Hard Limit** fields to 0, which indicates no limit.
5. Select **Add**.

Add a user quota onto a quota tree

Create a user quota on a quota tree to limit or track the amount of storage space that individual users consume on that tree. When you create user quotas on a tree, you can to use the default grace period and default hard and soft limits that are set at the tree-quota level.

Steps

1. Select **Storage** > **File** > **File Systems** > **[file system]** > **Quotas** > **Tree Quotas**.
2. Select a path, and click **Add User Quota**.
3. On the **Add User Quota** screen, provide the requested information. To track space consumption without setting limits, set the **Soft Limit** and **Hard Limit** fields to 0, which indicates no limit.

Index

A

audience 3

C

comments 3
conventions for publication 3

P

preface 3

R

related documentation 3

S

support information 3